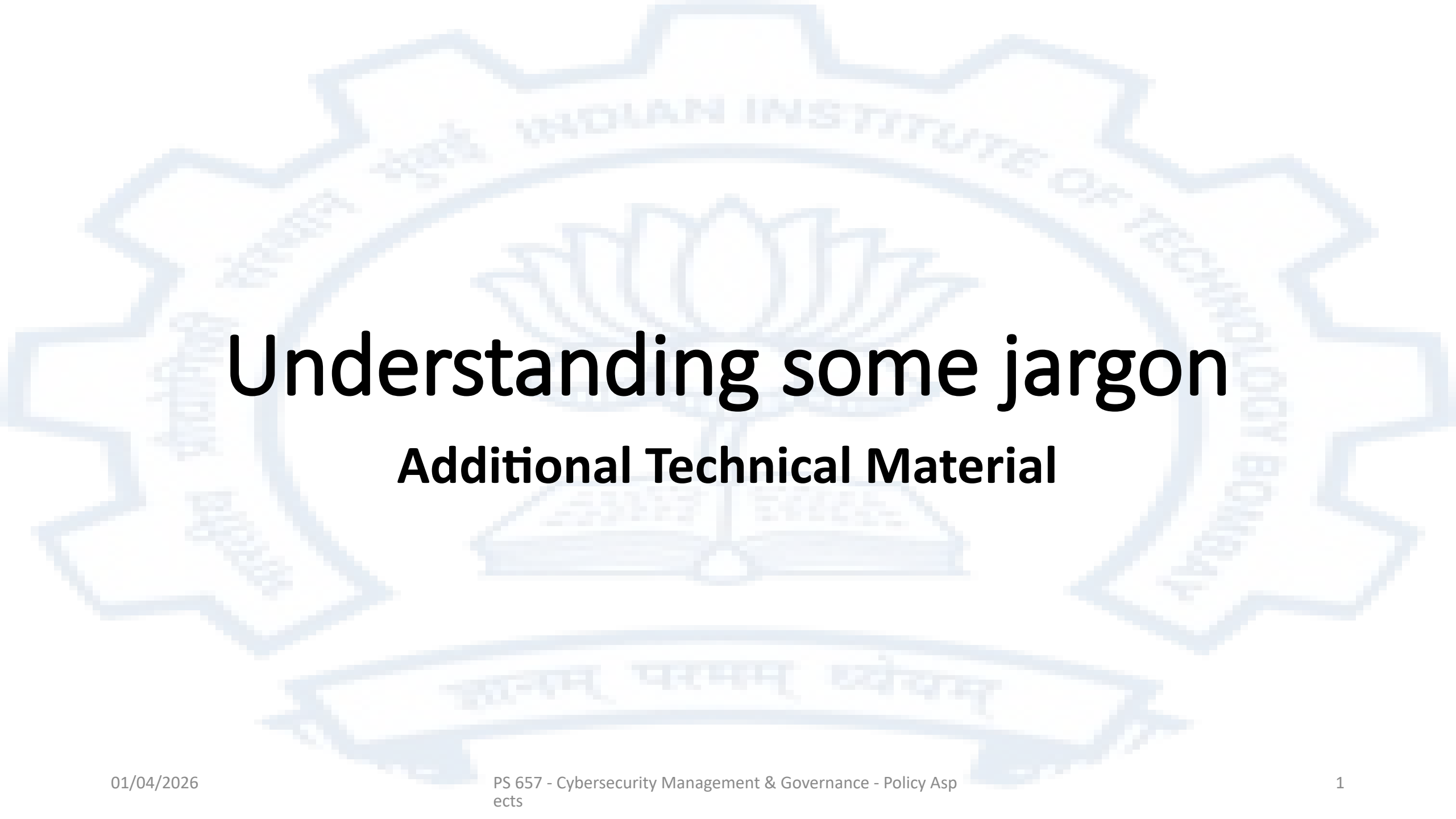


The background of the slide features a large, light blue watermark of the Indian Institute of Technology Bombay logo. The logo is circular with a gear-like outer border. Inside the circle, there is a lotus flower in the center, and the text "INDIAN INSTITUTE OF TECHNOLOGY BOMBAY" is written in a circular path around the lotus. At the bottom of the logo, there is a banner with text in Sanskrit.

Understanding some jargon

Additional Technical Material

EDR systems (Endpoint Detection and Response) evolving to XDR (Extended Detection and Response)

- The earliest antivirus systems were standalone systems. Each instance had its own signature database and was unaware of anything else. They did have the capability to update the database
- As the next step there was the ability to share information amongst a group of instances and this sharing delivered value
- Now there is the ability to share with designated global repositories in addition to closed user groups as well as run ML based classifiers to detect malware. These systems now have the capability to responding to threats as well. Some of the leading EDR solutions today are
- Crowdstrike Falcon, Palo Alto Cortex, Microsoft Defender, Sentinel One Singularity and Trend Micro Vision One

Zero Trust

- Conceptual move away from traditional perimeter based security which was based on an “inside” and an “outside”. Each device is assumed to be possibly compromisable and there for the is no implicit trust and continuous verification. That is the theory. In practice this takes two forms
- Microsegmentation of networks – in the extreme case each device is a distinct network
- Fine grained authentication. In the context of layered APIs each layer requires authentication.